ADITYA UNIVERSITY

(Formerly known as Aditya Engineering College (A))

000404

PART-I

S.No: 421944

Programe :	B.TECH
Hall Ticket No:	22A91A0334
Name :	PALAPARTHI KRISHNA BABU
Examination :	III SEMESTER SUPPLEMENTARY (AR20(R))
Month-Year :	Nov/Dec 2025
Branch :	MECHANICAL ENGINEERING
Course Code :	201BS3T12
Course Name :	Integral Transforms And Applications Of PartialDifferential Equations
Date of Exam :	18-11-2025

J. Pavan.

Signature of the Controller of Examination

18-11-25.

Signature of the Student with date

Signature of the Invigilator with date

Serial No. Last Page Written	16.

INSTRUCTIONS TO STUDENTS

1. The Answer Booklet contains 36 pages Ensure all the 36 pages are in proper order.
2. Student must verify the details of particulars in the PART - 1 i.e Name, Hall Ticket No., Examination, Course Name etc.,
3. In case of any deviation in the above or if the PART-1 is torn / damaged, report to the invigilator and return the damaged booklet.
4. You are prohibited from writing on or tampering the PART-1 except affixing the signature and serial number of last page written in the space provided.
5. Student is prohibited from:
   i. Writing their Hall Ticket No. and name in any part of the answer booklet.
   ii. Addressing the examiner in any manner whatsoever in the answer booklet. If they do so, their script will not be valued.
   iii. Writing religious symbols.
   iv. Either seeking or providing any assistance to the fellow students in the examinations.
   v. Possessing a manuscript or a printed matter, in any form, in the examination hall.
   vi. Bringing Mobile Phones / Cameras/ Bluetooth Devices / Programmable calculators etc.
   vii. Violation of the above mentioned instructions will be viewed as a case of malpractice, which is punishable offence.
6. Before beginning to answer any question, the students should write the correct number of that question in the margin provided. Answers written at different places for the same question will not be valued.
7. Students should write the answers, within the margins provided on both sides of the paper and on all the lines of each page. It is not necessary to begin each answer in a fresh page. Answers must be legibly written with blue or black pen.
8. Do not write anything except Question Number in the margin.
9. No loose sheets of papers will be allowed in the examination hall. No paper must be detached from or attached to the answer booklet except graph sheet.
10. Strike off all unused pages.
11. **NO ADDITIONAL ANSWER BOOKLETS/SHEETS WILL BE SUPPLIED.**

SPACE FOR ROUGH WORK

Q.No.

UNIT-I.

1(a) Fourier Series of $f(x) = \begin{cases} x & \text{if } 0 \le x \le 3 \\ 6-x & \text{if } 3 \le x \le 6. \end{cases}$

Fourier Series.

$$F(x) = \frac{a_0}{n} + \sum_0^{\infty} a_n \cos nx + \sum_0^{\infty} b_n \sin nx$$

where

$$a_0 = \frac{1}{\pi}\int_0^3 f(x)\,dx$$

$$a_n = \frac{1}{\pi}\int_0^3 f(x) \cos nx \, dx$$

$$b_n = \frac{1}{\pi}\int_0^3 f(x) \sin nx \, dx.$$

$$a_0 = \frac{1}{\pi}\int_0^3 f(x)\,dx = \int_0^3 f\left(\frac{1}{x}\right) dx$$

$$\frac{1}{\pi} = \int_0^3 f\left(\frac{\cancel{x}+1}{n+1}\right) dx$$

$$\frac{1}{\pi} = \int_0^3 f(\cancel{x}) = 6.$$

$$a_0 = \frac{6}{\pi}$$

$$uv = uv_1 - \int u' v_2 - \int u'' v_3$$

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$$a_n = \frac{1}{\pi}\int_0^3 f(x)\cos nx\, dx$$

$$\frac{1}{\pi}\int_0^3 f(x) \frac{-\sin nx}{n+1} dx$$

$$\frac{1}{\pi}\int_0^3 f(\cancel{x}) = \frac{-\sin nx}{n+1} dx$$

$$\frac{1}{\pi} - \int_0^3 \frac{-\sin 6x, \cos 6x}{\frac{6+1}{6+1}}$$

$$a_n = \frac{1}{\pi} \cancel{\quad} \int_0^3 \frac{-\sin 6x, \cos n x}{\frac{n+1}{n+1}}$$

$$a_n = 2\pi.$$

$$b_n = \frac{1}{\pi}\int_0^3 f(x)\sin nx\, dx$$

$$= \frac{1}{\pi}\int_0^3 f(x)\sin 6x\, dx$$

$$= \frac{1}{\pi}\int_0^3 f(x) \frac{-\cos 6x}{n+1} dx$$

$$= \frac{1}{\pi}\int_0^3 f(x) \frac{3\pi}{6+1} = \frac{3\pi}{7}$$

Q.No.

$b_n = 1/2\pi$.

from eq ① substitute the value.

$$\frac{a_0}{n} + \sum_0^{\infty} a_n \cos n x + \sum_0^{\infty} b_n \sin n x .$$

$$f(x) = \frac{6}{\pi} + \sum_0^{\infty} 2\pi \cos n x + \sum_0^{\infty} 1/2\pi \sin n x$$

$$= \frac{6}{\pi} + \sum_0^{\infty} 2\pi \cos 6x + \sum_0^{\infty} 1/2\pi \sin 6x$$

$$\frac{6}{\pi} + \sum_0^{\infty} \frac{(2\pi - \sin 6x)}{6x} + \sum_0^{\infty} \frac{1 - \cos 6x)}{6-1}$$

$$\frac{6}{\pi} - \frac{2\pi - \sin 6x}{6x} - \frac{(1 - \cos 6x)}{5} \, dx$$

Q.No.

1(b) fourier series of $f(x) = x\cos x$ in the interval $(-\pi, \pi)$.

Given that $f(x) = x\cos x$

we know that

$$f(x) = \frac{a_0}{n} + \sum_{0}^{\infty} a_n \cos nx + \sum_{0}^{\infty} b_n \sin nx$$

where $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx.$$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} x\cos x\,dx$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} x - \sin nx\,dx$$

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$$= \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{-\sin nx}{n+1} \, dx$$

$a_0 = 2.$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos x \quad \cos nx \, dx$$

$$\frac{1}{\pi} \int_{-\pi}^{\pi} \frac{-\sin nx + \cos nx}{n+1} \, dx$$

$$\frac{1}{\pi} \int_{-\pi}^{\pi} = (\sin nx + \cos nx)_{-\pi}^{\pi}$$

$$\left(\frac{\sin nx + \cos nx}{\pi} \right)_{-\pi}^{\pi}$$

$$a_n = \frac{2\pi}{n}.$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos x \quad \sin nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^{\pi} \sin x - \cos nx \, dx$$

Q.No.

In the equation

$$f(x) = \frac{a_0}{n} + \sum_0^\infty a_n \cos nx + \sum_0^\infty b_n \sin nx .$$

$$\cancel{=} f(x) = 1 + \sum_0^\infty 2\pi \cos nx + \sum_0^\infty 2\pi \sin nx$$

$$= 1 + \sum_0^\infty 2\pi \cos nx \, dx + \sum_0^\infty \pi \sin nx \, dx$$

$$2\pi \frac{-\sin dx}{n} + \sum_0^\infty \pi \frac{-\cos nx}{n+1}$$

Q.No.

Unit - III.

(5a) To Find $L\{\int_0^t e^{-t} \cos t \, dt\}$.

we know That

$$L\{\int_0^t e^{-t} \cos t \, dt\}.$$

$$\cdot\ i = f(t) = f(\tau)\int_0^{t} e^{t} f(\tau) f(t)$$

$$L\{\int_0^t e^{-t} \cos t \, dt\}$$

$$f(t) = \int_0^t e^{-t} \cos t \, dt)$$

$$f(\tau)\int_0^t e^{t} f(\tau) f(t)$$

$$f(\tau)\int_0^t e^{-t} \cos t \, dt \ f(t)$$

$$f(x)\int_0^t e^{t} \frac{\sin nx \ dt \ f(t)}{\cos nx \ dt \ f(\tau)}.$$

$$f(n) \int_0^t e^{t} \ \frac{-\sin x}{\cos x} \ \frac{f(t)}{f(\tau)}.$$

$$\hookrightarrow \int_0^t e^{t} \ \frac{-\sin x}{\cos x}$$

Q.No.

$$= \int_0^{\infty} e \frac{\cos 6t - \cos ut}{t}$$

$$= \{t^2 e^{-3t} \sin 2t\}$$

$$= e^t \cos 2n x\ 2t\ f(s)$$

$$= f(n) = f(s) = f(t)$$

$$f(t) = x \cos x$$

Q.No.

5(b) To find $L\{f(t)\}$ where $f(t) = \begin{cases} k \text{ if } 0<t<a \\ -k \text{ if } a<t<2a. \end{cases}$

Given that

$L\{f(t)\}$ where $f(t) = \begin{cases} k \text{ if } 0<t<a \\ -k \text{ if } a<t<2a. \end{cases}$

$L = f(t) = f(s) \int_0^1 e^t f(s) f(t).$

$L\{f(t) = \{ k$

$f(t) = \int_0^1 e^t f(s) f(k)$

$\int_0^1 e^t f(s) = \cos nx \; f(k)$

$\int_0^1 e^t f(\sin nx) = \cos nx \; f(\bar{k}).$

$\int_0^1 e^t (f \sin nx) = f(\bar{k}).$

$= L\{f(t)\}$ where $f(t) = \sin x \, dx$

$\int_0^1 e^t f(s) f(t)$

Q.No.

Unit - II .

3(a) Using Fourier integral,

Show that $\int_0^\infty \frac{\cos \lambda x}{1+\lambda^2} d\lambda \, e^{-x}, x \geq 0.$

The given

$$\int_0^\infty \frac{\cos \lambda x}{1+\lambda^2} d\lambda \, e^{-x}, x \geq 0.$$

$$\int_0^\infty \frac{\cos \lambda x}{1+\lambda^2} d\lambda \, e^{x}$$

$$\int_0^\infty \frac{-\sin n x \, \lambda x}{1+\lambda^2} \int_0^\infty e^{-x}$$

$$= \int_0^\infty \frac{-\sin n x \, d x}{1+\lambda^2} \int_0^\infty e^{-x}$$

$$= \int_0^\infty \frac{\cos \lambda x}{1+\lambda^2} d\lambda \, e^{x}$$

$$= \int_0^\infty \frac{\cos 1 x}{1+1^2} d\lambda$$

$$= \int_0^\infty \frac{\cos 1 x}{2^2} d\lambda$$

Q.No.

$$\int_0^\infty \frac{\cos\lambda x}{1+\lambda^2} \int_0^\infty e^{-x}$$

$$= \int_0^\infty \frac{\cos\lambda x}{1+\lambda^2}\, d\lambda\ e^{-x},\ x \geq 0.$$

$$= \int_0^\infty \left\{ \frac{1}{(1+\lambda^2)^2}\, d\lambda\ e^{-x}.\right.$$

Q.No.

Unit-IV.

7a) $L^{-1}\left\{\frac{2s-3}{s^2+4s+13}\right\}$

$L^{-1}\left\{\frac{2s-3}{s^2+4s+13}\right\}$

$L^{-1}\left\{\frac{2s-3}{s^2+4s+13}\right\}$

$\Rightarrow L^{-1}\frac{2s}{s^2+4s+13} \quad L^{1}\frac{-3}{s^2+4s+13}$

$\frac{2s}{s^2+4s+13} + \frac{-3}{s^2+4s+13}$

$= \frac{2s+n+1}{s^2+4s+13} + \frac{-3n+1}{s^2+4s+13}$

$= \frac{2s+2+1}{\cancel{8s}\,\cancel{6}17s^2} + \frac{-3n+1}{17s^2}$

$= \frac{2s+2+1-3n+1}{17s^2}$

$= \frac{\cancel{4}\,5s-3n+1}{17s^2}$

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$$\frac{5S - 3\cancel{S} + 1}{17S^2}$$

$$= \frac{2\cancel{S} + 1}{17\cancel{S}}$$

$$= \frac{3}{17}$$

$$= 0.17647$$

7b) Using convolution theorm.

find $L^{-1}\left\{\frac{s^2}{(s^2+4)(s^2+9)}\right\}$

In convolution theorm

$$f(x) = f(\bar{a}) + f(\bar{b}) \cdot dx$$

$$L^{-1}\left\{\frac{s^2}{(s^2+4)(s^2+9)}\right\}$$

$$\frac{s^2}{(s^2+4)} = a \quad \frac{s^2}{(s^2+9)} = b$$

$$f(x) = f\left(\frac{s^2}{(s^2+4)}\right) + f\left(\frac{s^2}{s^2+9}\right)\right) dx$$

Q.No.

$$f(x)\ \frac{s^2}{(s^2+4)} + \frac{s^2}{(s^2+9)}\Big)$$

$$= \left(\frac{s^2+s^2}{s^2+4+s^2+9}\right)$$

$$= \left(\frac{2s^2}{2s^2+13}\right)$$

$$= \left(\frac{2s^2+s^2}{(s^2+9)+(s^2+4}\right)$$

$$L^{-1}\left\{\frac{s^2}{(s^2+4)(s^2+9)}\right\}$$

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SEMI-LOG PAPER (5 CYCLES X 1/10")